

Gyroscope on Aeroplane

by

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What is Gyroscope?

Definition:

A gyroscope is a device for measuring or maintaining orientation, based on the principles of the conservation of angular momentum.

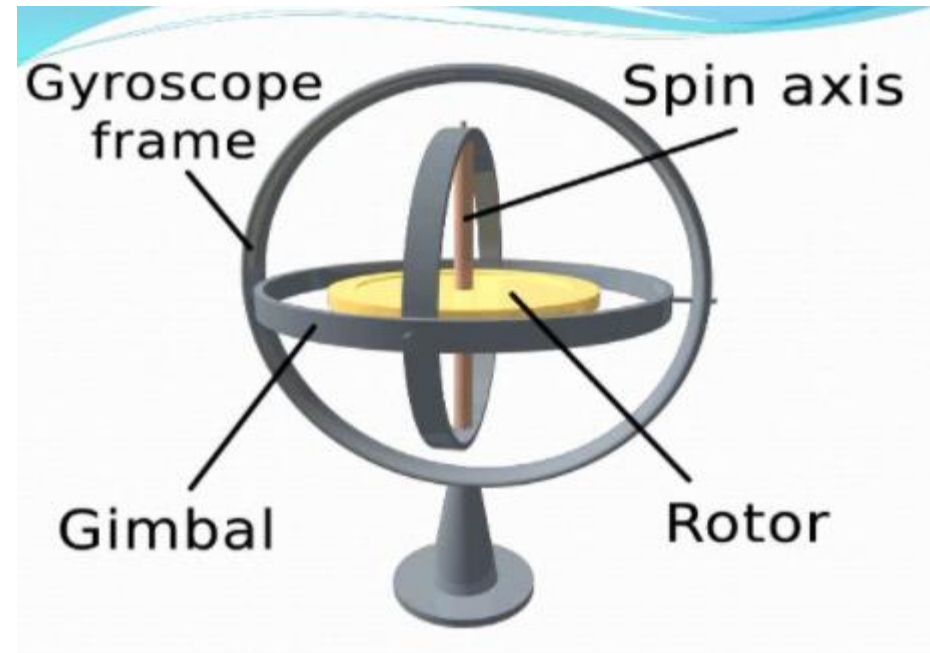


Fig. 1. Gyroscope

- If the axis of spinning or rotating body is given an angular motion about an axis perpendicular to the axis of spin, an angular acceleration acts on the body about the third perpendicular axis.
- The torque required to produce this acceleration is known as the active gyroscopic torque.
- A reactive gyroscopic torque or couple also acts similar to the concept of centripetal and centrifugal forces on a rotating body.
- The effect produced by the reactive gyroscopic couple is known as the gyroscopic effect.
- Thus aeroplane, ships, automobiles, etc, that have rotating parts in the form of wheels or rotors of engines experiences gyroscopic effect while taking turn, such as when axes of spin is subjected to some angular motions.

Gyroscope contd...

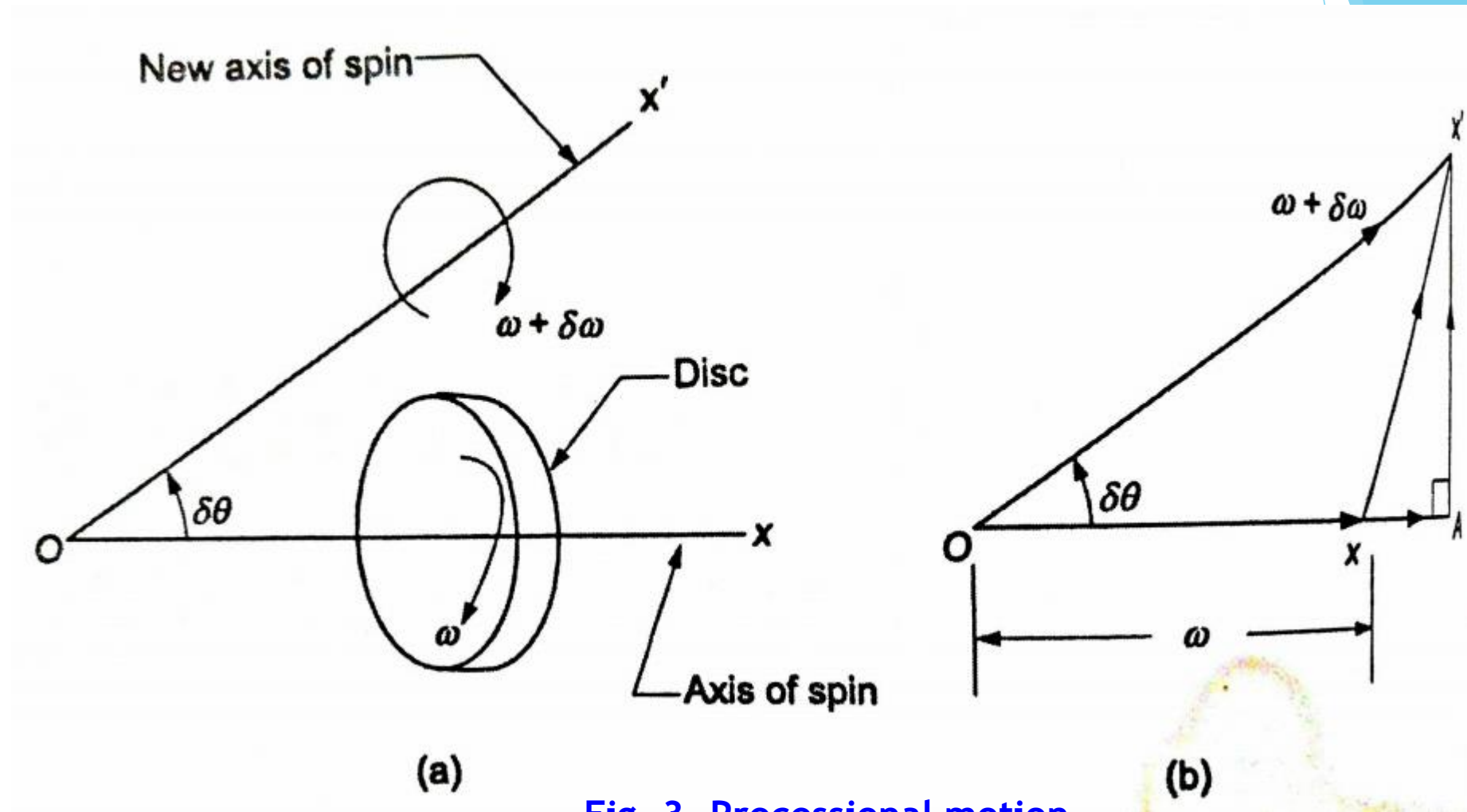


Fig. 3. Precessional motion

Gyroscope contd...

Angular acceleration along

$$\begin{aligned} OX &= \lim_{\delta t \rightarrow 0} \left[\frac{(\omega + \delta\omega) \cos \delta\theta - \omega}{\delta t} \right] \\ &= \lim_{\delta t \rightarrow 0} \left[\frac{\omega \cos \delta\theta + \delta\omega \cos \delta\theta - \omega}{\delta t} \right] \\ &= \lim_{\delta t \rightarrow 0} \left[\frac{\omega + \delta\omega - \omega}{\delta t} \right] \quad (\text{Because } \cos \delta\theta \approx 1) \\ &= \lim_{\delta t \rightarrow 0} \frac{\delta\omega}{\delta t} \\ &= \frac{d\omega}{dt} \end{aligned}$$

Gyroscope contd...

Angular (or gyroscopic) acceleration perpendicular to OX ,

$$\begin{aligned} &= \lim_{\delta t \rightarrow 0} \left[\frac{(\omega + \delta\omega) \sin \delta\theta}{\delta t} \right] \\ &= \lim_{\delta t \rightarrow 0} \frac{(\omega \cdot \delta\theta + \delta\omega \cdot \delta\theta)}{\delta t} \\ &= \frac{\omega \cdot d\theta}{dt} \quad [\text{Neglecting } \delta\omega \cdot \delta\theta, \text{ being small}] \\ &= \omega \cdot \omega_p \end{aligned}$$

where ω_p is the precessional angular speed of the spin axis.
Total angular acceleration of disc = XX'



Gyroscope contd...

$$\begin{aligned} &= XA \quad AX' \\ &= \frac{d\omega}{dt} \quad \omega \cdot \frac{d\theta}{dt} \\ &= \frac{d\omega}{dt} \quad \omega \cdot \omega_p \end{aligned}$$

where $\rightarrow$ represents vector sum.

Gyroscope contd...

Total angular acceleration along XX' may be written as:

$$\alpha = \frac{d\omega}{dt} + \omega \frac{d\theta}{dt}$$

1. $d\omega/dt$, represents change in the magnitude of the angular velocity of the rotor.
2. $\omega.d\theta/dt$, represents change in the direction of the axis of spin, the direction of AX' is from A to X' in the vector diagram (being a component of XX'), the acceleration acts clockwise in the vertical plane xz (when viewed from front along y -axis).

Gyroscope contd...

Fundamentals of Gyroscopic Motion:

Gyroscope: It is a body which spinning about an axis is free to move in other direction under the action of external forces.

Axis of Spin: It is axis about which the body revolves.

Gyroscopic effect: consider a body spinning about an axis OX . If couple represented by vector OZ perpendicular to OX .

- Then the body tries to precess about an axis OY , which is perpendicular to both OX and OZ .
- This combined effect is called gyroscopic or precessional effect.
- The plane of spin, plane of precession, and plane of gyroscopic couple are mutually perpendicular.

Gyroscope contd...

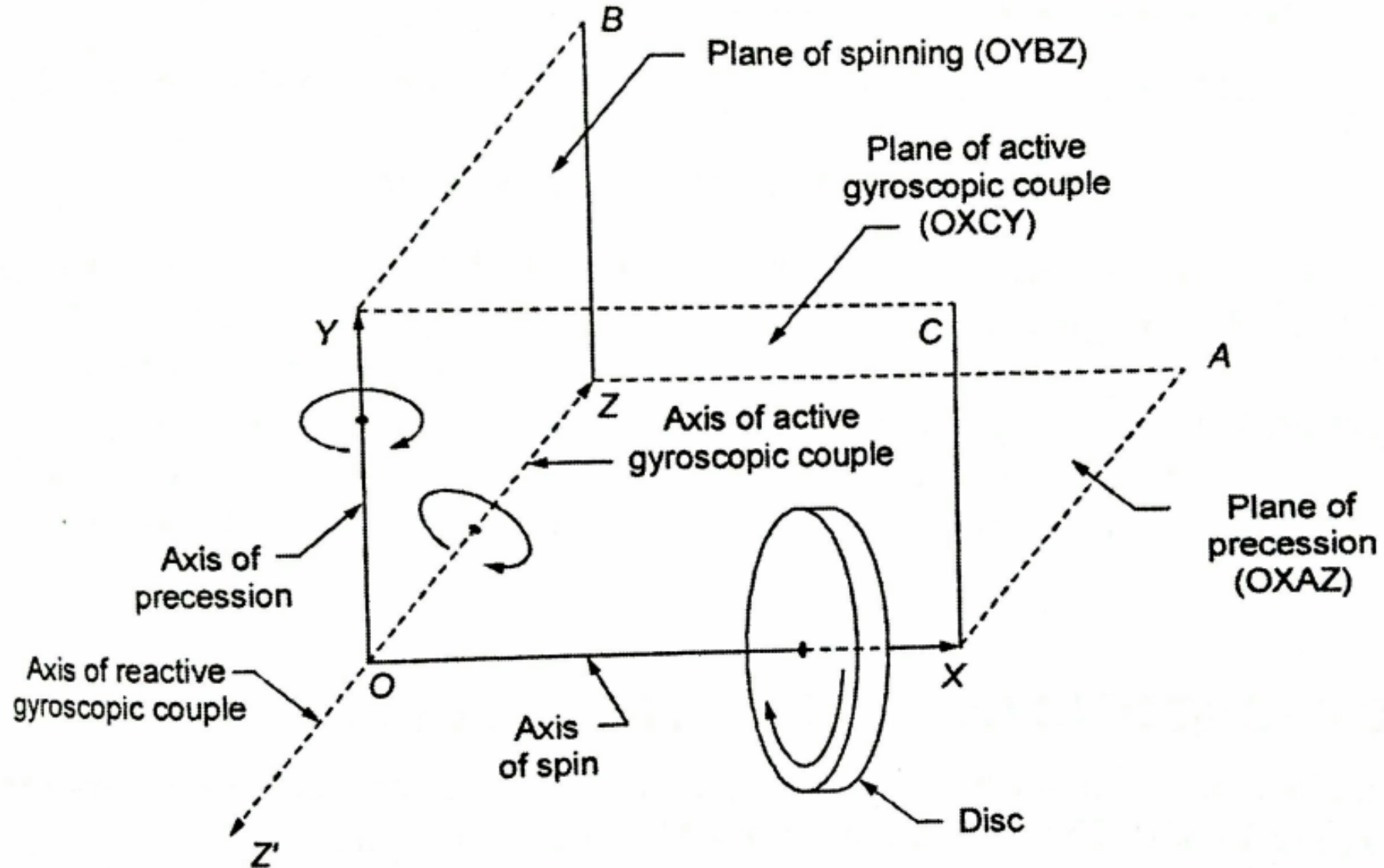


Fig. 4. Gyroscopic effect

Gyroscope contd...

Gyroscopic Couple:

Torque or Couple C:

$$T = I \frac{d\omega}{dt} + T\omega \frac{d\theta}{dt}$$

$$T = I\alpha + T\omega\omega_p$$

$$\alpha = \frac{d\omega}{dt} + \omega \frac{d\theta}{dt}$$

Gyroscopic Couple

- Aero planes are subjected to gyroscopic effect when it taking off, landing and negotiating left or right turn in the air.

- Let

ω = Angular velocity of the engine rotating parts in rad/s,

m = Mass of the engine and propeller in kg,

k = Radius of gyration in m,

I = Mass moment of inertia of engine and propeller in kg m²,

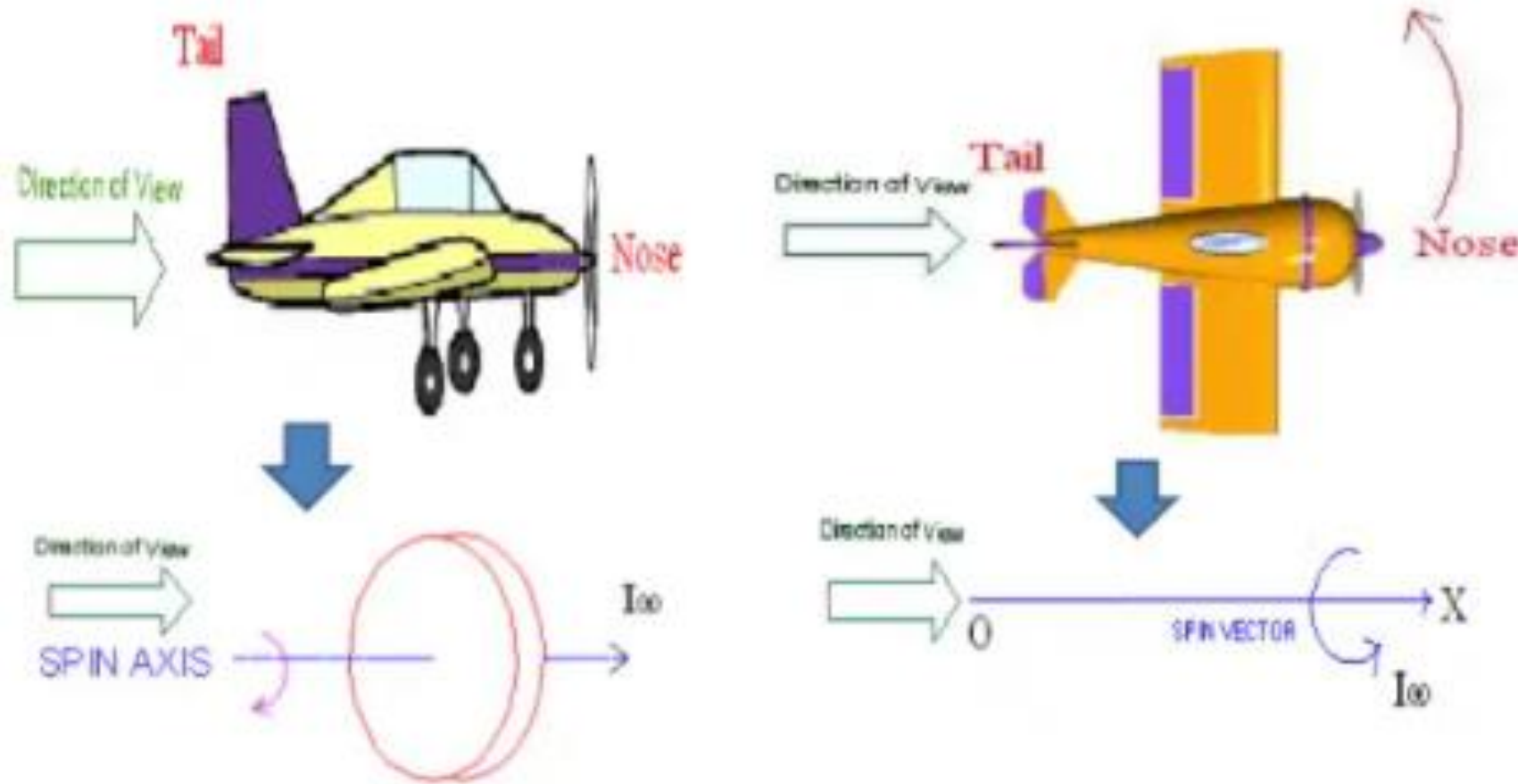
V = Linear velocity of the aero plane in m/s,

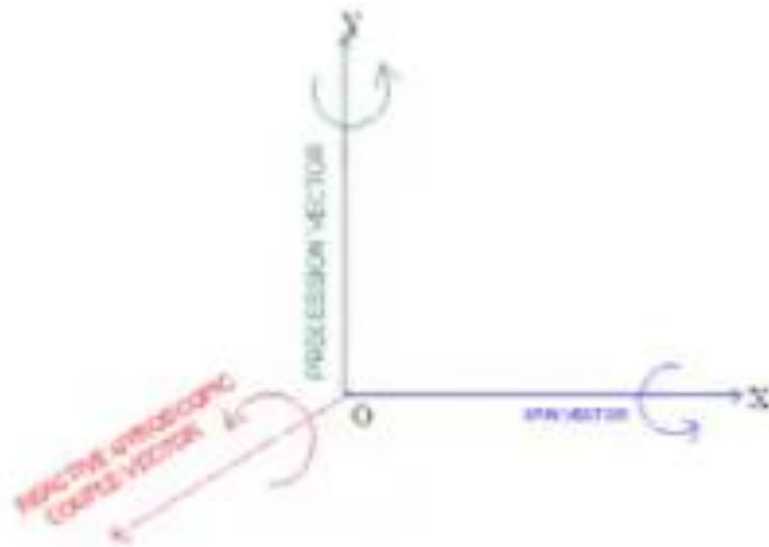
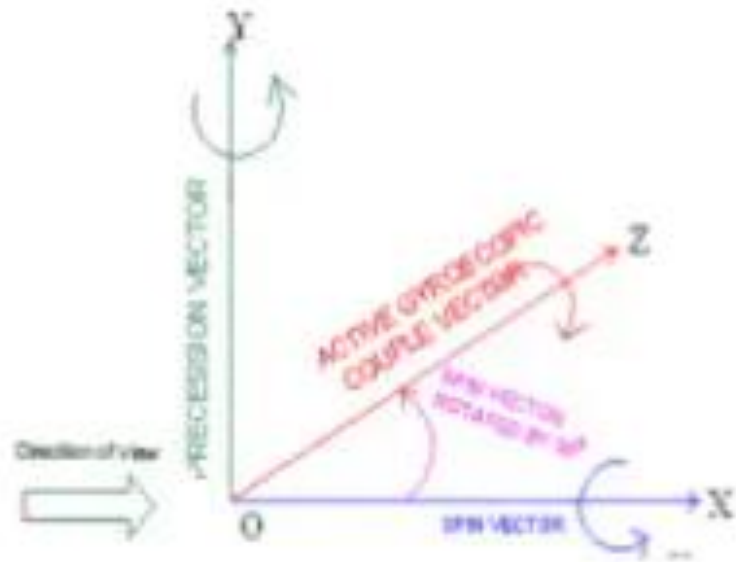
R = Radius of curvature in m,

ω_p = Angular velocity of precession = rad/s

Gyroscopic couple acting on the aero plane = $C = I \omega \omega_p$

Case (i): PROPELLER rotates in CLOCKWISE direction when seen from rear end and Aeroplane turns towards LEFT

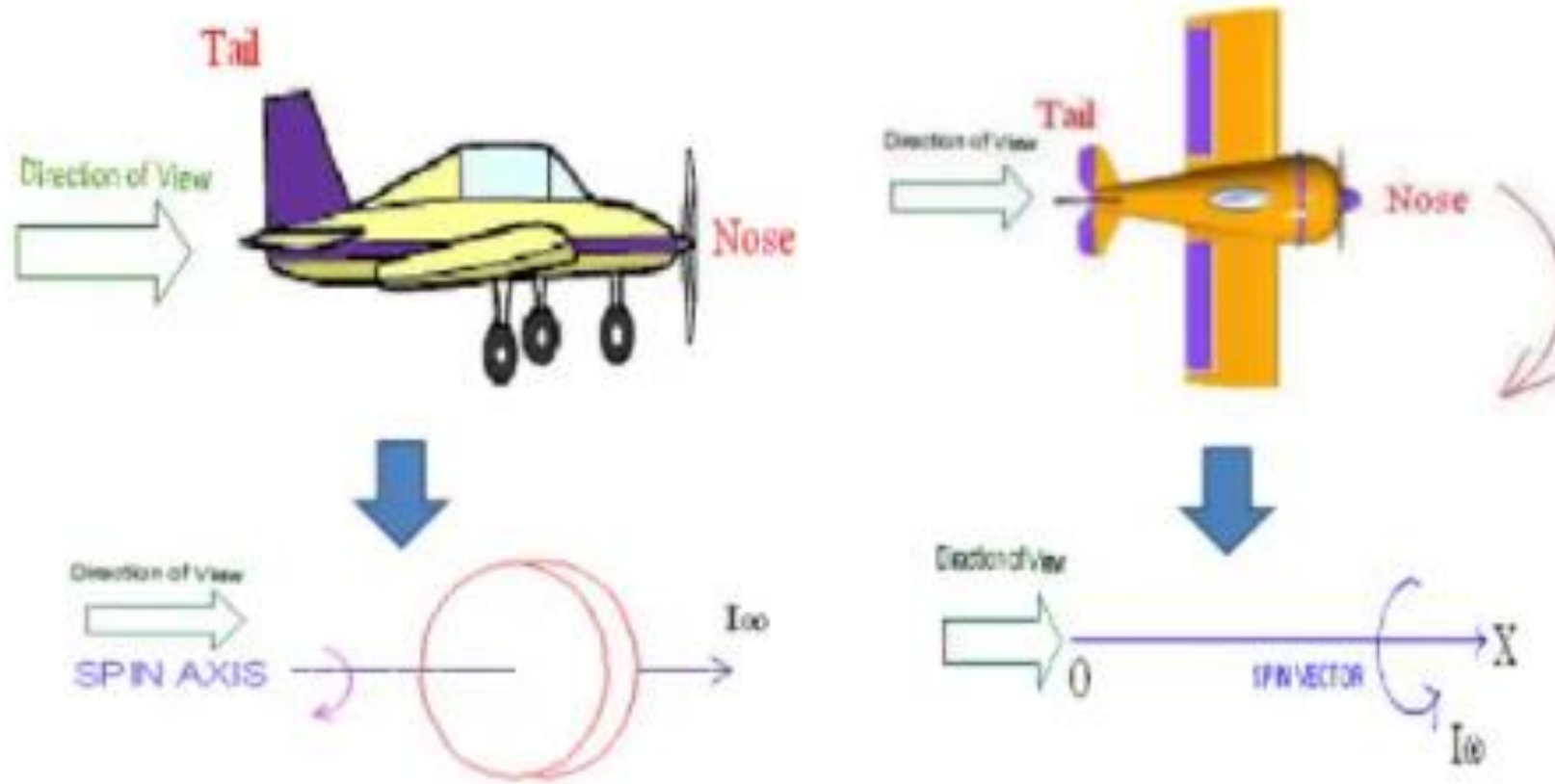




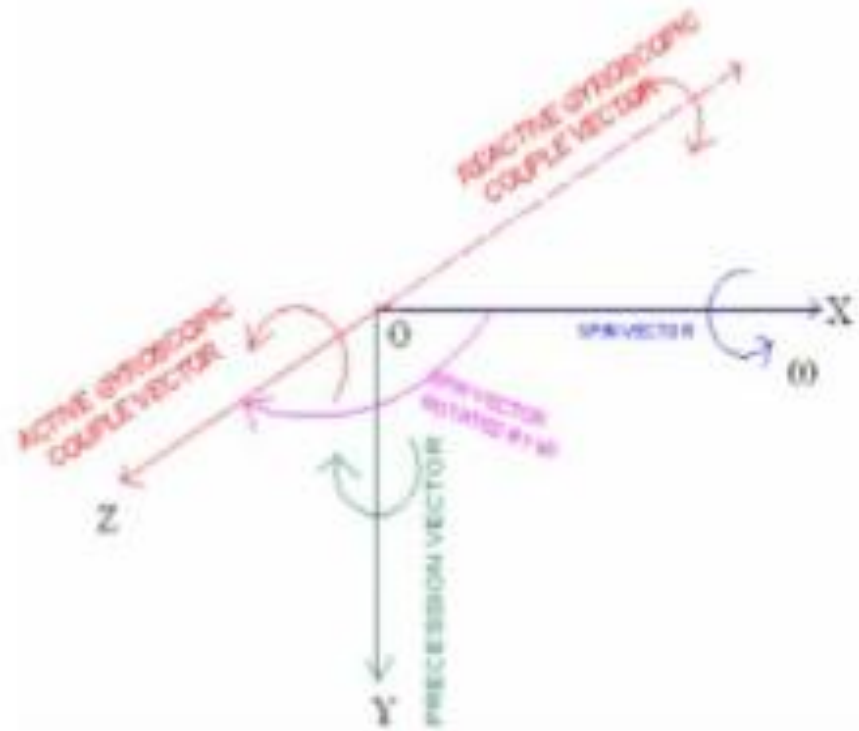
- According to the analysis, the reactive gyroscopic couple tends to dip the tail and raise the nose of aeroplane.



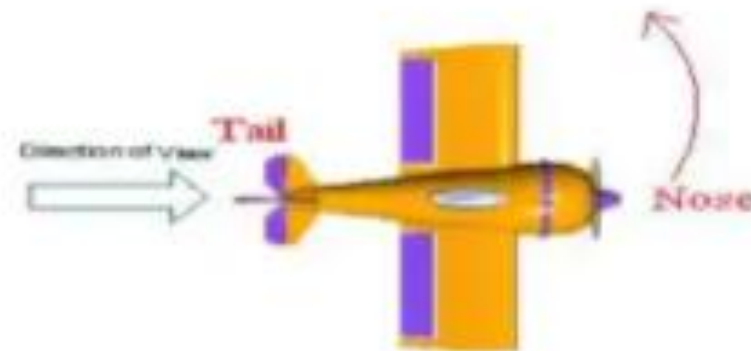
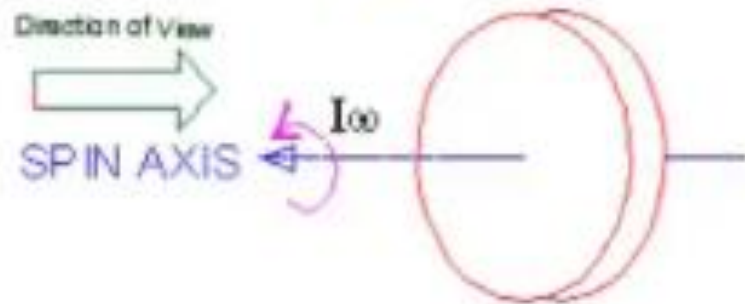
Case (ii): PROPELLER rotates in CLOCKWISE direction when seen from rear end and Aeroplane turns towards RIGHT



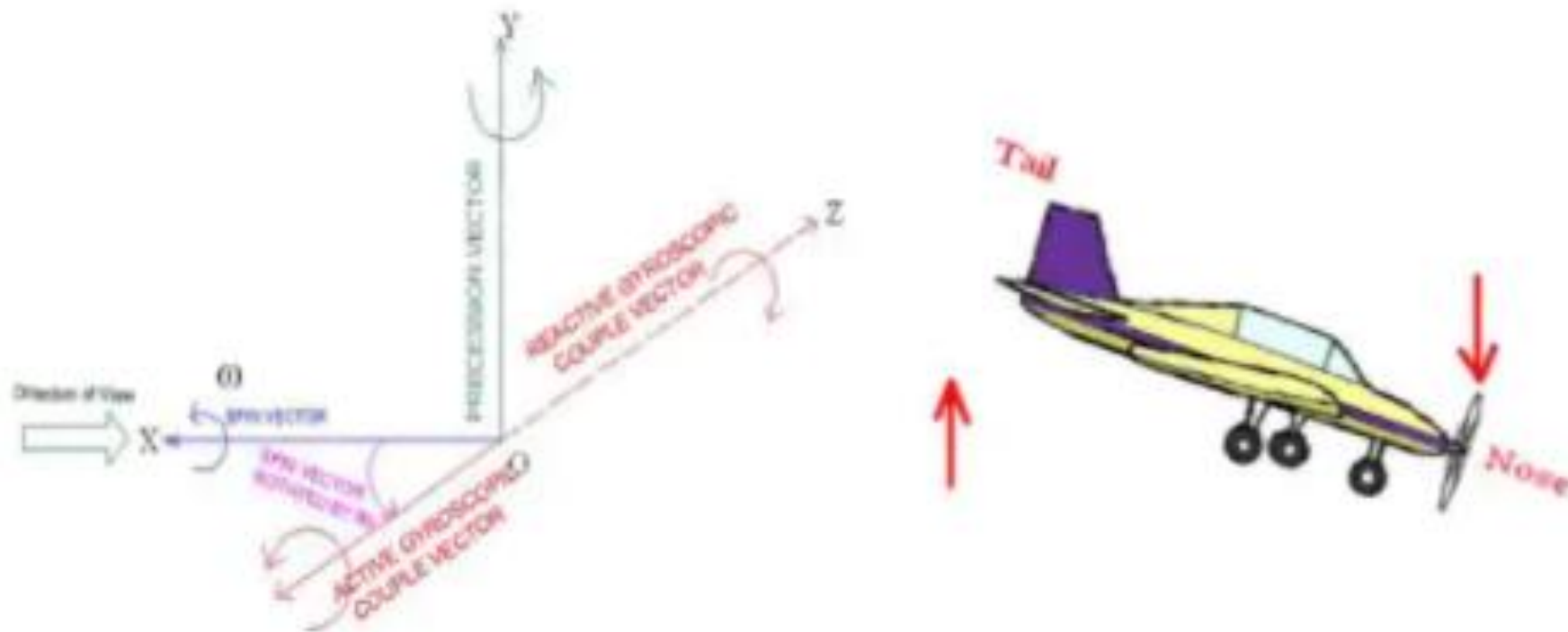
- According to the analysis, the reactive gyroscopic couple tends to raise the tail and dip the nose of aeroplane.



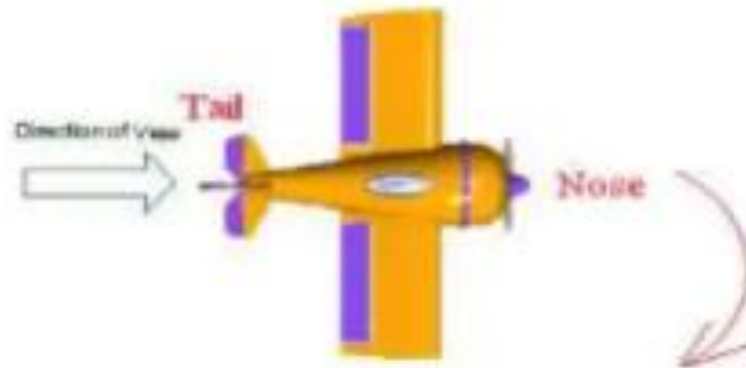
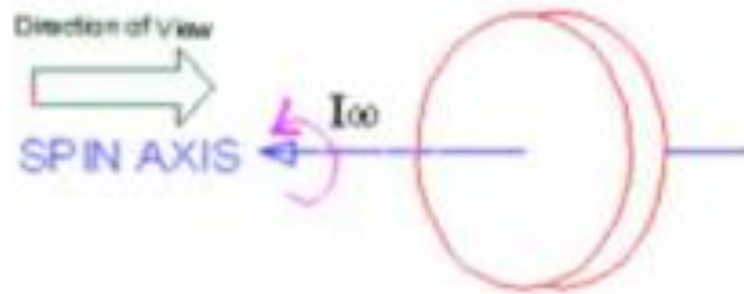
Case (iii): PROPELLER rotates in ANTICLOCKWISE direction when seen from rear end and Aeroplane turns towards LEFT



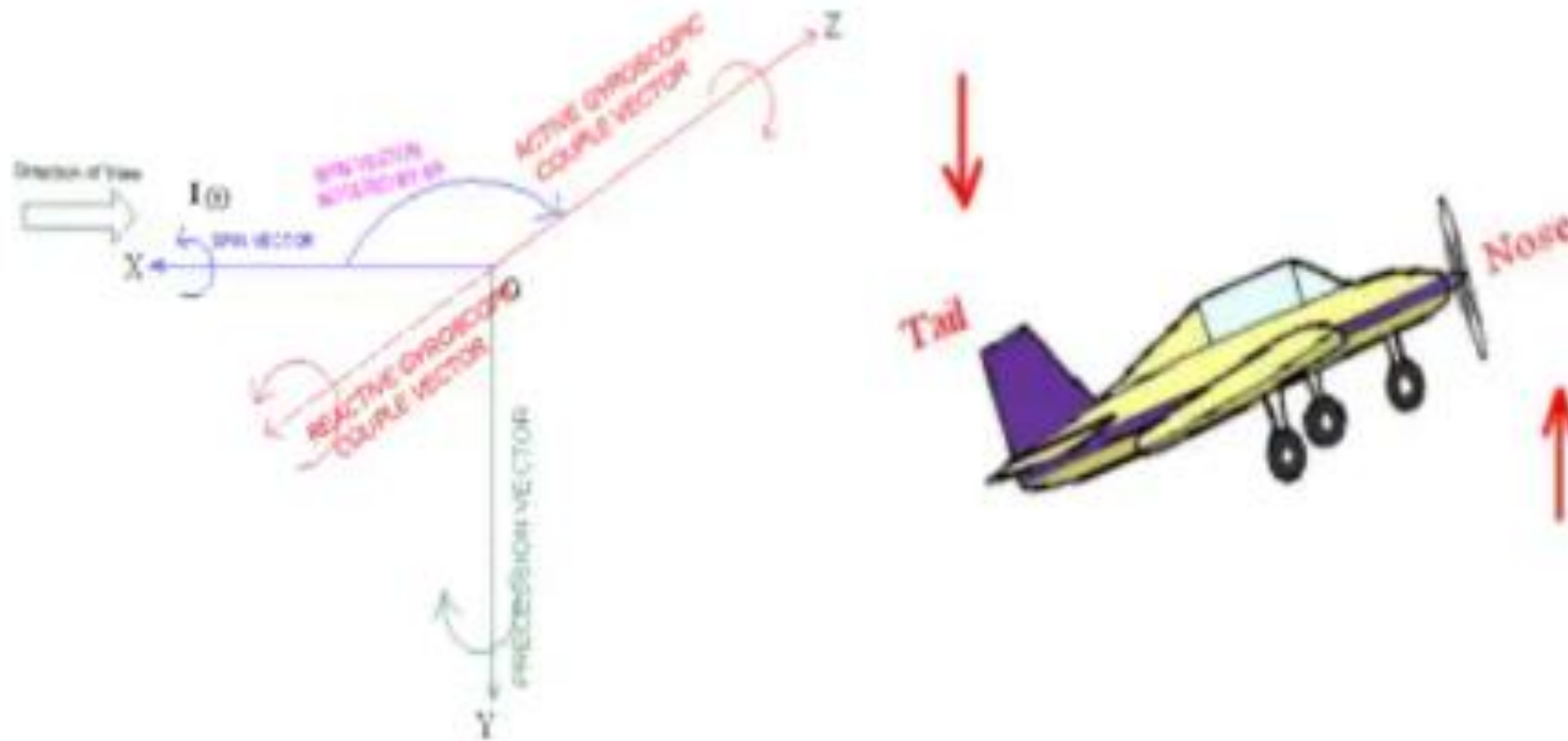
- According to the analysis, the reactive gyroscopic couple tends to raise the tail and dip the nose of aeroplane



Case (iv): **PROPELLER** rotates in **ANTICLOCKWISE** direction when seen from rear end and Aeroplane turns towards **RIGHT**



- The analysis shows, the reactive gyroscopic couple tends to raise the tail and dip the nose of aeroplane



Example 17.3 *An aeroplane flying at 240 km/h turns towards the left and completes a quarter circle of 60 m radius. The mass of the rotary engine and the propeller of the plane is 450 kg with a radius of gyration of 320 mm. The engine speed is 2000 rpm clockwise when viewed from the rear. Determine the gyroscopic couple on the aircraft and state its effect.*



In what way is the effect changed when the

- (i) aeroplane turns towards right*
- (ii) engine rotates clockwise when viewed from the front (nose end) and the aeroplane turns (a) left (b) right?*

Solution

$$m = 450 \text{ kg}, k = 0.32 \text{ m},$$

$$\omega = \frac{2\pi \times 2000}{60} = 209.4 \text{ rad/s},$$

$$v = \frac{240 \times 10^3}{3600} = 66.67 \text{ m/s}$$

$$I = mk^2 = 450 \times (0.32)^2 = 46.08 \text{ kg.m}^2$$

$$\omega_p = \frac{v}{r} = \frac{66.67}{60} = 1.11 \text{ rad/s}$$

$$C = I \omega \omega_p = 46.08 \times 209.4 \times 1.11 \\ = 10\,713 \text{ N.m} = 10.713 \text{ kN.m}$$

The effects of gyroscopic couples on the aeroplane when it takes a left or right turn are discussed in the previous section. In brief it is summarized below:

Figure 17.6(a) shows the aeroplane in space. When it turns left,

oa is the angular momentum vector before turning [Fig. 17.6(b)];

ob is the angular momentum vector after turning;

ab is the applied or active gyroscopic couple;

b'a' is the reaction couple and is perpendicular to **oa** in the limit. This couple acts in the vertical plane and tends to raise the nose and depress the tail of the aeroplane.

(i) If the aeroplane takes a right turn, the reaction couple is **d'e'** [Fig. 17.6(c)], which shows that the nose is depressed and the tail is raised.

(ii) (a) When the engine rotates or spins clockwise on viewing from the nose end, **ef** is the applied couple as the aeroplane turns left [Fig. 17.6(d)]. **f'e'** is the reaction couple indicating that the nose is depressed and the tail is raised.

(b) When the aeroplane takes a right turn, the tail is depressed and the nose is raised [Fig. 17.6(e)].

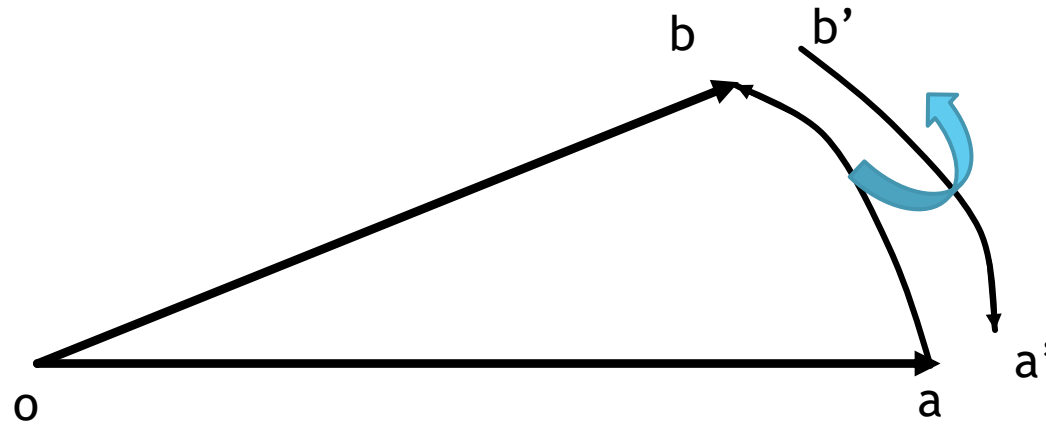


Fig. Aeroplane takes left turn and engine is rotating clockwise viewer is seen form rear side or from tail side

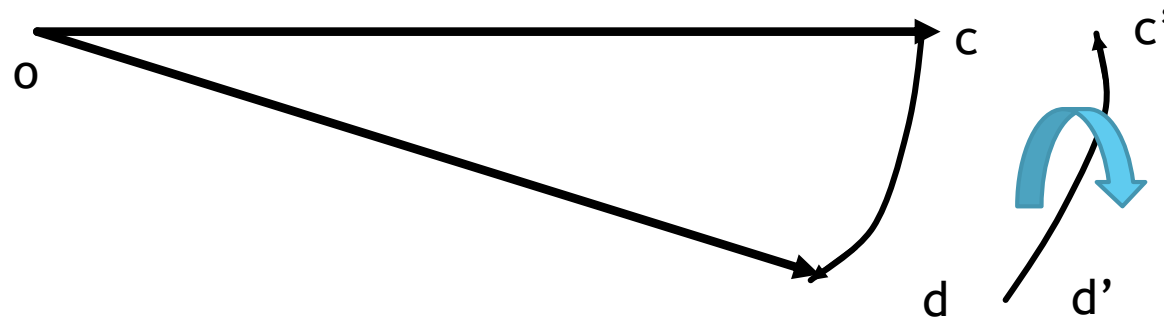


Fig. Aeroplane takes right turn and engine is rotating clockwise viewer is seen form rear side or from tail side

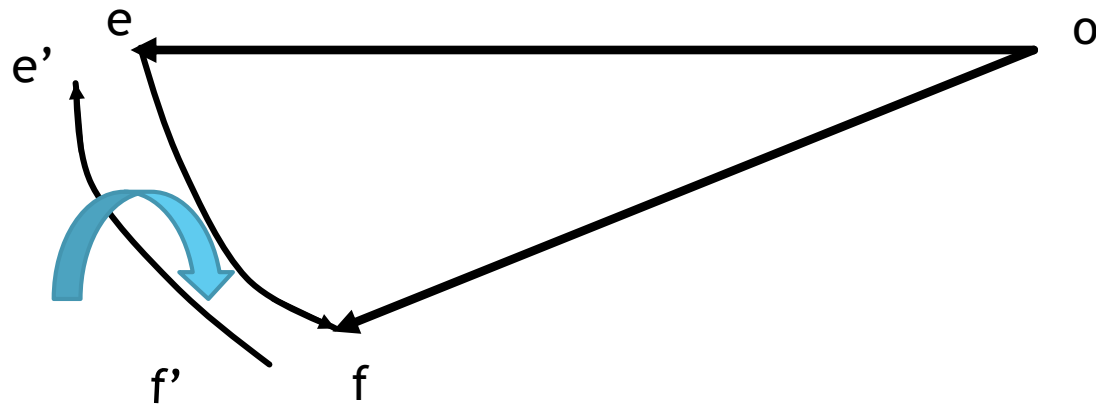


Fig. Aeroplane takes left turn and engine is rotating clockwise viewer is seen from front side or nose side

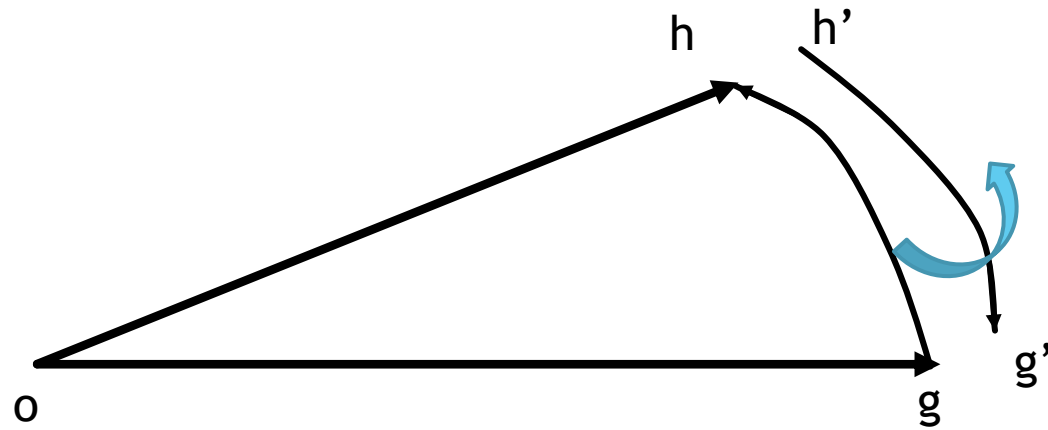


Fig. Aeroplane takes right turn and engine is rotating clockwise viewer is seen from front side or nose side

Thank You

